

Design

Nsenda SM 225103522

Input

For each superhuman

Name

Power

ID

For each hero

Number of captured villains

For each villain

Amount of money stolen

Number of people injured

Output

For each superhuman

Name

Power

ID

For each hero

Rating

Number of captured villains

For each villain

Amount of money stolen

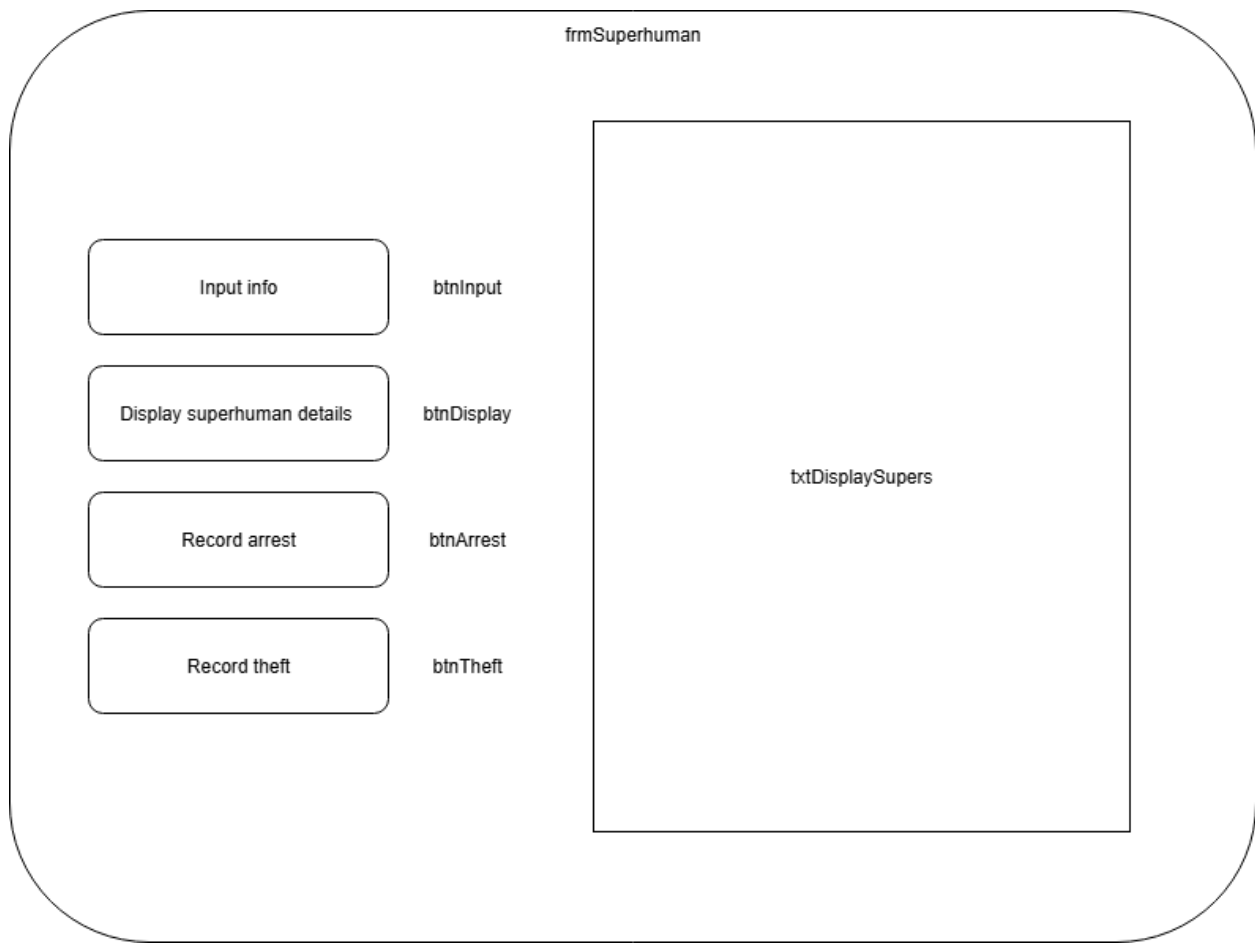
Number of people injured

Events & actions

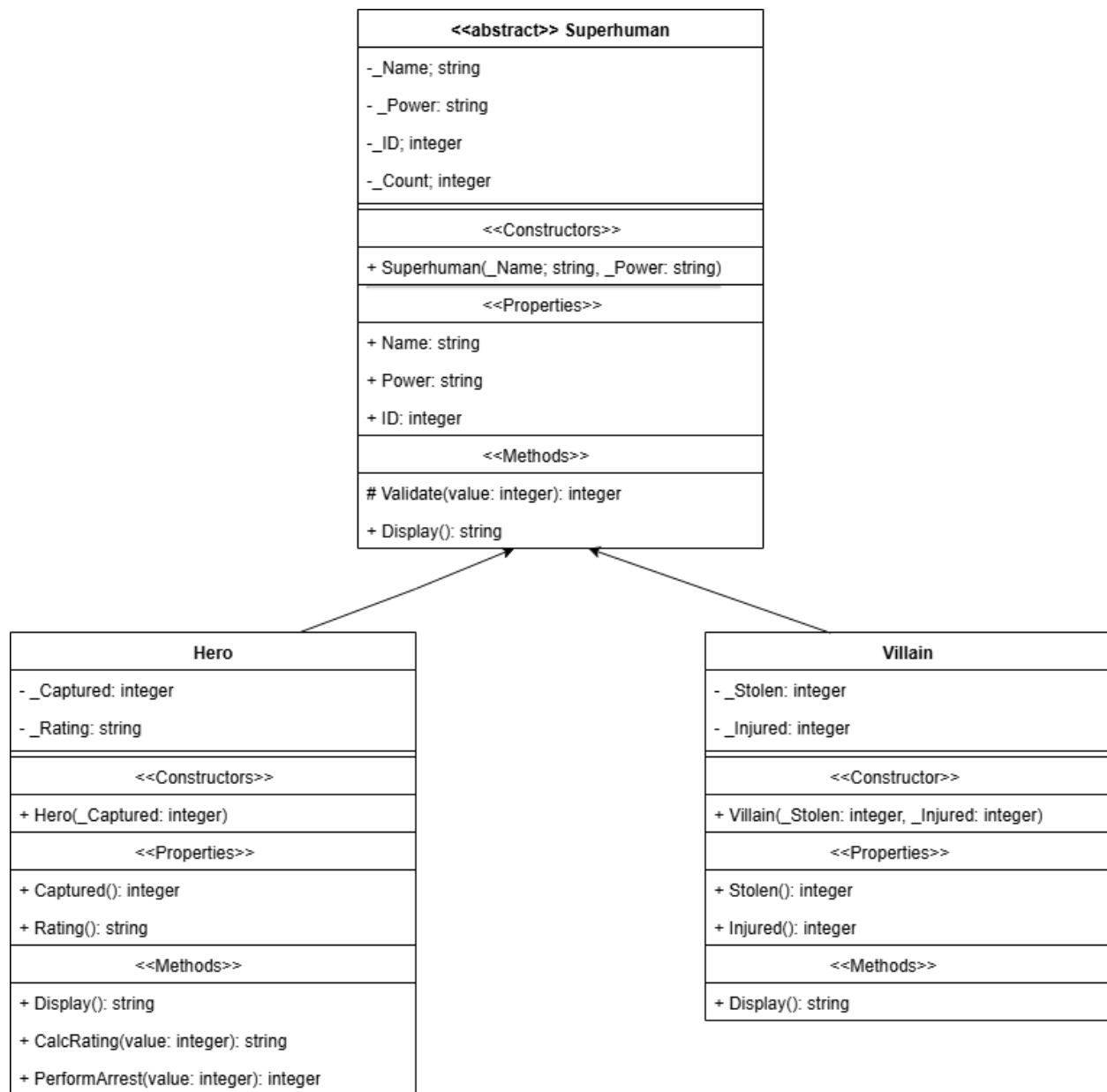
Events	Actions
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User clicks btnInput	Get number of superhumans Resizes array Instantiate objects
User clicks btnDisplay	Display super human's details
User clicks btnArrest	Record number of arrests Recalculates rating Update superhuman display
User clicks btnTheft	Record number of people hurt Record total amount stolen Update superhuman display

Interface



UML



Variables

NumSupers as integer

Superhumans() as Superhuman

Algorithm

1. NumSupers <- Inputbox
2. Resize Superhumans array
3. Get superhuman type from user
4. Name, power <- inputbox
5. Instantiate object for different superhumans
6. Perform upcasting: superhuman(s) <- hero/villain
7. txtDisplaysupers.text <- superhuman(s).display()
8. get super human's id
9. downcast object
10. record thefts/arrests, id it downcasts
11. don't record anything if you can't downcast
12. update display

Test Data

# Super humans	3		
Type	Hero	Villain	Hero
ID	1	2	3
Name	Flash	Poison Ivy	Beast Boy
Power	Speed	Nature	Shapeshift
# Captured	5		49
Rating	Rookie		Competent
Money stolen		1000	
People injured		3	
Record arrest	5+15=20		49+2=51
	Competent		Super hero
Record theft		1000+500=1500	
		3+1=4	